CS4011 — Principles of Machine Learning

Tier	A
Phase (order)	5
Do this first	CS6015 or equivalent, CS2800
Estimated effort	12-14 weeks, ~9 h/week
Unlocks	CS5691, CS6910, CS6700, CS6370, CS6350

Roadmap

1. Linear/ridge regression; Bayesian classification and parameter estimation.
2. Evaluation and model selection; kNN; logistic regression, perceptron, SVM.
3. PCA/LDA; decision trees; ensembles; clustering.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- NPTEL MOOC Machine Learning 2016 (88 videos). NPTEL *Introduction to Machine Learning* (Balaraman Ravindran, IIT Madras): this is the IITM ML course.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Stanford CS229: Machine Learning led by Andrew Ng | Autumn 2018 (21 videos). Stanford CS229 Autumn 2018 (Andrew Ng, official Stanford Online): more maths-heavy; use the free CS229 notes.
- Machine Learning Course - CS 156 (18 videos). Caltech CS156 *Learning from Data* (Abu-Mostafa, official): the best explanation of generalisation/VC theory.
- Machine Learning Foundations (93 videos). IIT Madras BS *Machine Learning Foundations*: maths refresh (linear algebra, PCA, probability) for ML.

⚡ Short videos for intuition and revision

- Machine Learning (100+ videos). StatQuest *Machine Learning*: 10-20 min intuitive explanation of each algorithm.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–5, 85–88	Intro, supervised/unsupervised/RL; the probability and linear algebra tutorials at the end of the playlist.

Week	Lectures	What to do
2	6–10	Decision theory, bias-variance, linear and multivariate regression. Implement linear regression with the normal equation and with gradient descent.
3	11–20, 84	Subset selection, ridge/lasso, PCR, PLS, logistic regression, LDA.
4	21–26, 83	Perceptron, SVMs, kernels, hinge loss. Use sklearn SVC on a real dataset; plot decision boundaries.
5	27–33	Neural networks + backprop, MLE, MAP, Bayesian estimation. Implement a 2-layer MLP in NumPy.
6	34–47, 82	Decision trees, evaluation measures, cross-validation, ROC. Implement a decision tree from scratch.
7	48–53	Hypothesis testing and confidence intervals (for comparing models).
8	54–58, 65	Bagging, boosting, gradient boosting, random forests, naive Bayes, multi-class. Kaggle Titanic with XGBoost.
9	59–64	Bayesian networks, undirected models, HMMs, variable elimination, belief propagation.
10-11	66–77	Clustering (k-means, hierarchical, BIRCH, CURE, DBSCAN), GMMs and EM, spectral clustering, frequent itemsets. Implement k-means and EM for a GMM.
12	78–81	Learning theory, intro to RL.

▼ **All lectures in *NPTEL MOOC Machine Learning 2016* (88 videos)**

1. Introduction to Machine Learning
2. Week 1 - Lecture 1 - Introduction to Machine Learning
3. Week 1 Lecture 2 - Supervised Learning
4. Week 1 Lecture 3 - Unsupervised Learning
5. Week 1 Lecture 4 - Reinforcement Learning
6. Week 2 Lecture 5 - Statistical Decision Theory - Regression
7. Week 2 Lecture 6 - Statistical Decision Theory - Classification
8. Week 2 Lecture 7 - Bias - Variance
9. Week 2 Lecture 8 - Linear Regression
10. Week 2 Lecture 9 - Multivariate Regression
11. Week 3 Lecture 10 Subset Selection 1
12. Week 3 Lecture 11 Subset Selection 2
13. Week 3 Lecture 12 Shrinkage Methods
14. Week 3 Lecture 13 Principal Components Regression
15. Week 3 Lecture 14 Partial Least Squares
16. Week 3 Lecture 15 Linear Classification
17. Week 3 Lecture 16 Logistic Regression
18. Week 3 Lecture 17 Linear Discriminant Analysis 1

19. Week 3 Lecture 18 Linear Discriminant Analysis 2
20. Week 3 Lecture 19 Linear Discriminant Analysis 3
21. Week 4 Lecture 20 Perceptron Learning
22. Week 4 Lecture 21 SVM - Formulation
23. Week 4 Lecture 22 SVM - Interpretation & Analysis
24. Week 4 Lecture 23 SVMs for Linearly Non Separable Data
25. Week 4 Lecture 24 SVM Kernels
26. Week 4 Lecture 25 SVM - Hinge Loss Formulation
27. Week 5 Lecture 26 ANN I - Early Models
28. Week 5 Lecture 27 ANN II - Backprogpogation I
29. Week 5 Lecture 28 ANN III - Backpropogation II
30. Week 5 Lecture 29 ANN IV - Initialization, Training & Validation
31. MAXIMUM LIKELIHOOD ESTIMATE
32. Week 5 Lecture 31 Parameter Estimation II - Priors & MAP
33. Week 5 Lecture 32 Parameter Estimation III - Bayesian Estimation
34. Week 6 Lecture 33 Decision Trees - Introduction
35. Week 6 Lecture 34 Regression Trees
36. Week 6 Lecture 35 Stopping Criteria & Pruning
37. Week 6 Lecture 36 Decision Trees for Classification - Loss Functions
38. Week 6 Lecture 37 Decision Trees - Categorical Attributes
39. Week 6 Lecture 38 Decision Trees - Multiway Splits
40. Week 6 Lecture 39 Decision Trees - Missing Values, Imputation & Surrogate Splits
41. Week 6 Lecture 40 Decision Trees - Instability, Smoothness & Repeated Subtrees
42. Week 6 Lecture 41 Decision Trees - Example
43. Week 6 Lecture 42 Evaluation Measures 1
44. Week 6 Lecture 43 Bootstrapping & Cross Validation
45. Week 6 Lecture 44 - 2 Class Evaluation Measures
46. Week 6 Lecture 45 - The ROC Curve
47. Week 6 Lecture 46 - Minimum Description Length & Exploratory Analysis
48. Week 7 Lecture 47 - Introduction to Hypothesis Testing
49. Week 7 Lecture 48 - Basic Concepts
50. Week 7 Lecture 49 - Hypothesis Testing II - Sampling Distributions & The Z test
51. Week 7 Lecture 50 -STUDENT'S T-TEST
52. Week 7 Lecture 51 - Hypothesis Testing IV - The Two Sample and Paired Sample t-tests
53. Week 7 Lecture 52 - Confidence Intervals
54. Week 8 Lecture 53 - Ensemble Methods - Bagging, Committee Machines and Stacking
55. Week 8 Lecture 54 - Boosting
56. Week 8 Lecture 55 - Gradient Boosting
57. Week 8 Lecture 56 - Random Forests
58. Week 8 Lecture 57 - Naive Bayes
59. Week 9 Lecture 58 Bayesian Networks

60. Week 9 Lecture 59 Undirected Graphical Models - Introduction
61. Week 8 Lecture 60 Undirected Graphical Models - Potential Functions
62. Week 9 Lecture 61 Hidden Markov Models
63. Week 9 Lecture 62 Variable Elimination
64. Week 9 Lecture 63 Belief Propagation
65. Lecture 64 Multi-class Classification
66. Week 10 Lecture 65 Partional Clustering
67. Week 10 Lecture 66 Hierarchical Clustering
68. Week 10 Lecture 67 Threshold Graphs
69. Week 10 Lecture 68 The BIRCH Algorithm
70. Week 10 Lecture 69 The CURE Algorithm
71. Week 10 Lecture 70 Density Based Clustering
72. Week 11 Lecture 71 Gaussian Mixture Models
73. Week 11 Lecture 72 Expectation Maximization
74. Week 11 Lecture 73 Expectation Maximization Continued
75. Lecture 76 Spectral Clustering
76. The Apriori Property
77. Frequent Itemset Mining
78. Lecture 79 Learning Theory
79. Lecture 80 Introduction to Reinforcement Learning
80. Lecture 81 - RL Framework and TD Learning
81. Lecture 82 Solution Methods & Applications
82. Week 6 Decision Trees Tutorial
83. Week 4 Tutorial 4 - Optimization
84. Week 3 Weka Tutorial
85. Week 2 Tutorial 2 - Linear Algebra (2)
86. Week 2 Tutorial 2 - Linear Algebra (1)
87. Week 1 Tutorial 1 - Probability Basics (2)
88. Week 1 Tutorial 1 - Probability Basics (1)

Books, notes and websites

- ✅ NPTEL *Introduction to Machine Learning* (Balaraman Ravindran, IIT Madras) — nptel.ac.in/courses/106106139/
- Stanford CS229 (Andrew Ng) — cs229.stanford.edu
- Book: Bishop, *Pattern Recognition and Machine Learning* (the course text, free PDF)
- Book: *An Introduction to Statistical Learning* (statlearning.com, free)
- scikit-learn tutorials; Kaggle Learn

Practice / proof of understanding

Re-implement linear regression, logistic regression, k-means and a decision tree from scratch with NumPy, then compare with scikit-learn on 3 Kaggle datasets.

What your IITM course sheet says

From the PDF: CS4011

Official description: This course will serve as a comprehensive introduction to various topics in machine learning. At the end of the course the students should be able to design and implement machine learning solutions to classification, regression, and clustering problems; and be able to evaluate and interpret the results of the algorithms.

Official syllabus (trimmed):

> - Introduction to the course, recap of linear algebra and probability theory basics.
> - Linear Regression, Ridge Regression, Sensitivity Analysis, Multivariate Regeression
> - Bayesian Classification: Naive Bayes, Parameter Estimation (ML, MAP), Sequential Pattern Classification
> - Evaluation and Model Selection: ROC Curves, Evaluation Measures, Significance tests
> - Non-parametric Methods: k-Nearest Neighbours, Parzen Window
> - Discriminative Learning models: Logistic Regression, Perceptrons, Artificial Neural Networks, Support Vector Machines
> - Dimensionality Reduction: Principal Component Analysis, Fischer's Discriminant Analysis
> - Decision Trees: Splitting Criteria, CART
> - Ensemble Methods: Boosting, Bagging, Random Forests
> - Clustering: Partitional, Hierarchical, density based clustering.

Official textbooks: 1. Pattern Recognition and Machine Learning, Christopher Bishop, Springer 2006.

Official references: 1. Introduction to Statistical Learning, Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Springer, 2013. 2. Pattern Classification, 2nd Ed., Richard Duda, Peter Hart, David Stork, John Wiley & Sons, 2001.

From the PDF: CS5691

Official description: At the end of the course, the students should be able to design and implement machine learning techniques for solving pattern recognition tasks such as classification, regression, and clustering, and be able to evaluate and interpret the results of the techniques.

Official syllabus (trimmed):

1. Basics of Linear Algebra, Probability Theory and Optimization: Vectors, Inner product, Outer product, Eigenanalysis, Probability distributions – Discrete distributions and Continuous distributions; Conditional probability distribution and Joint probability distribution, Bayes theorem, Unconstrained optimization, Constrained optimization – Lagrangian multiplier method. (5 Lectures)
2. Methods for Function Approximation: Linear models for regression, Parameter estimation method, Regularization, Ridge regression (6 Lectures)
3. Probabilistic Models for Classification: K-Nearest neighbours classifier, Bayesian decision theory, Bayes classifier, Normal (Gaussian) density – Discriminant functions, Decision surfaces, Maximum-Likelihood estimation; Gaussian mixture models -- Expectation-Maximization method for parameter estimation; Naive Bayes classifier, Non-parametric techniques for density estimation -- Parzen-window method, K-nearest neighbors method (15 Lectures)
4. Discriminative Learning based Models for Classification: Logistic regression, Perceptron, Multilayer feedforward neural network – Gradient descent method, Error backpropagation method; Support vector machine. (12 Lectures)
5. Dimensionality Reduction Techniques: Principal component analysis, Fisher discriminant analysis (3 Lectures)
6. Non-Metric Methods for Classification: Decision trees, Random fore

Official textbooks: 1. C.M.Bishop, Pattern Recognition and Machine Learning, Springer, 2006 2. R.O.Duda, P.E.Hart and D.G.Stork, Pattern Classification, John Wiley, 2001

Official references: 1. S.Theodoridis and K.Koutroumbas, Pattern Recognition, Academic Press, 2009 2. E.Alpaydin, Introduction to Machine Learning, Prentice-Hall of India, 2010